

TRUDIE WANG, Ph.D.

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SUMMARY

Mission driven engineering and product leader with two decades of experience architecting distributed energy platforms, optimization engines, and intelligent control systems that accelerate the global transition to sustainable and equitable infrastructure. Deep expertise in DER optimization, microgrids, VPPs, forecasting, and distributed control, grounded in a lifelong commitment to resource democratization and systems level design. Recognized for unifying engineering, product, utilities, policymakers, financiers, and communities into aligned ecosystems. Known for infectious enthusiasm, cross domain fluency, and the ability to inspire teams, secure strategic partnerships, and represent organizations at the highest levels. Played a decisive role in the successful acquisition of climate tech startups through product vision, technical leadership, and high stakes stakeholder engagement. After three startups and years of navigating the trials, tribulations, and breakthroughs of scaling climate technologies, I am focused on applying the full breadth of my technical, product, business development, and mentorship experience to help organizations avoid repeated pitfalls and accelerate their impact. I am seeking product and technical leadership roles that allow me to build, guide, and scale transformative climate solutions at the level this critical moment for our planet demands.

CORE EXPERTISE

- Distributed optimization and control (MPC, ADMM)
 - Microgrids, VPPs, DERMS, transactive energy
 - Product strategy, roadmapping, market alignment
 - Forecasting, analytics, algorithm architecture
 - Systems thinking across energy, water, food, waste
 - Stakeholder engagement across utilities, developers, financiers, communities
 - Technical leadership, team building, mentorship
 - Public speaking, industry representation, thought leadership
 - Global deployments in LMI communities and developing countries
 - Sustainability, resource equity, circular systems
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PROFESSIONAL EXPERIENCE

HEILA TECHNOLOGIES (Acquired)

VP of Innovation | 2023-2024

VP of Product | 2021-2023

Product Strategy Lead | 2020-2021

Boston, MA / Oakland, CA

- Led innovation, product strategy, and distributed optimization for an MIT born microgrid and DER control platform acquired into the Kohler family. Unified forecasting, optimization, telemetry, and control into a scalable architecture deployed across community microgrids, resilience hubs, C&I campuses, and international pilots across Central and South America, Europe, Asia, and Australia.
- Played a decisive role in Heila's acquisition by defining and presenting the product roadmap, articulating the long term technical and market vision, and providing the domain expertise that enabled the company to pass due diligence and secure a strong valuation.
- Served as a primary external representative to investors, utilities, and strategic partners, using technical fluency and systems level storytelling to win major projects and build confidence among seed round investors and industry stakeholders.

- Led high impact deployments including EcoBlock, Red Cross resilience hubs, decentralized community microgrids with Emera Technologies in Tampa and PEPCO, and C&I and community projects across developed and developing countries.
- Directed algorithm and product teams to significantly improve optimization performance across all production sites, enabling value stacking, resilience services, and market participation.
- Unified engineering, product, BD, utilities, and community partners to ensure technology met real world constraints and long term market needs.
- Represented Heila at major industry conferences including Distributech, RE+, NABCEP, IEEE, Smart Grid, Wood Mackenzie, and Stanford, securing strategic partnerships and elevating brand visibility.
- Mentored engineers and product managers, cultivating a mission driven, diverse team with strong technical rigor and collaborative culture.
- Served as the Rialto Wastewater Treatment Plant subject matter expert and executive technical lead on behalf of Tablerock, guiding municipal partners, engineering firms, and investors through the design, modeling, and deployment of California's first microgrid-enabled wastewater treatment facility.
- Led Heila's internal technical strategy for strategic and complex multi-stakeholder deployments across infrastructure, C&I, and community microgrids, including DER modeling, digital twin development, tariff and market analysis, and coordination with municipal leadership and engineering partners.
- Acted as technical domain lead for NSF-funded research and national energy grants advancing distributed control, microgrid intelligence, and transactive energy architectures.
- Represented Heila in federal, state, and municipal grantmaking initiatives, translating Heila's microgrid IP into scalable frameworks for community microgrids, critical infrastructure, and C&I campuses.
- Drove cross-functional collaboration to mature Heila's microgrid IP into production-ready architectures deployed across global strategic pilots, NSF-funded research, and municipal and C&I projects.

GROWING ENERGY LABS INC. (Acquired)

Staff Analytics Engineer | 2017-2020

Senior Analytics Engineer | 2015-2017

San Francisco, CA

- Developed forecasting, optimization, and aggregation algorithms for early VPP architectures and DER control systems.
- Contributed to GELI's acquisition through technical leadership, customer engagement, and cross functional collaboration with utilities and international partners across Asia and Australia.
- Led ARPA E NODES projects and international DER pilots, including distributed optimization deployments in New Zealand.
- Designed aggregation architectures and control policies for multi asset DER portfolios, improving dispatch efficiency and grid support capabilities.

SOLARCITY

Power Systems Engineer | 2014-2015

San Francisco, CA

- Managed DER integration pilots and analyzed grid impacts of solar and storage systems.
- Developed models and analytics to accelerate renewable penetration and improve grid reliability.

ELECTRIC POWER RESEARCH INSTITUTE (EPRI)

Graduate Research Intern | 2010-2012

Palo Alto, CA

- Developed dynamic and distributed control algorithms for DERs using MPC and ADMM.

- Quantified load balancing potential and grid benefits of DER coordination using real world data.

SELECTED IMPACT PROJECTS

- EcoBlock community microgrid and resilience initiative in LMI neighborhoods
- Red Cross resilience hubs for emergency preparedness
- Emera Technologies decentralized community microgrids in Tampa
- PEPCO community microgrids for resilience and equity
- ARPA E NODES distributed optimization pilots in the US and New Zealand
- C&I deployments across industrial campuses globally
- Rialto Wastewater Treatment Plant microgrid (executive technical lead)
- NSF microgrid intelligence research and national energy grant programs

THOUGHT LEADERSHIP AND PUBLIC SPEAKING

Featured speaker, moderator, or panelist at:

Distributech • RE+ Grid Edge • NABCEP • IEEE Tech Expo • Smart Grid • Greentown Labs • Wood Mackenzie Grid Edge Innovation Summit • Stanford Graduate Energy Seminar • Environment + Energy Leader • Interchange Recharged Podcast • Authority Magazine Interview • International Microgrid Symposium • NY Climate Week (Capgemini Deep Tech Panel) • Microgrid Knowledge (moderator) • GridTECH (moderator)

EDUCATION

Stanford University – Ph.D., Mechanical Engineering

Renewable energy systems, distributed optimization, controls, DER integration

Stanford University – M.S., Mechanical Engineering

UC Berkeley – B.S., Mechanical Engineering (Honors)

PUBLICATIONS

Wang, O'Neill, Kamath. *Dynamic Control and Optimization of Distributed Energy Resources in a Microgrid*, IEEE Transactions on Smart Grid.

Wang, Kamath, Willard. *Control and Optimization of Grid Tied Photovoltaic Storage Systems using MPC*, IEEE Smart Grid Transactions.

Wang, Peumans. *Designing a Metallic Nanoconcentrator for a Multijunction PV Cell*, Journal of Applied Physics.

MISSION AND VALUES

Lifelong commitment to sustainability, resource democratization, and social equity.

Systems thinker who believes the energy transition must empower people and communities.

Known for “infect by example” leadership, living a low impact lifestyle and inspiring others through action, mentorship, and global engagement.